

SignSee

Cloud Deployment Cost Analysis

Research Group 2 • BUas Year 2 • June 2026

1. Purpose & Scope

This document evaluates the financial cost of SignSee's Azure cloud infrastructure, covering both actual costs incurred during development and projected costs for an independent production deployment. SignSee is a real-time Dutch Sign Language (NGT) fingerspelling recognition system consisting of a containerised FastAPI inference backend, a Vue 3 frontend, and a periodic ML retraining pipeline.

Costs are analysed across two dimensions: compute (the cost of running ML training jobs and inference) and deployment (the cost of the infrastructure that delivers the application to users). All figures are drawn from Azure Cost Management filtered to the team-r2-2026 resource, and the Azure Pricing Calculator for forward-looking projections.

2. Actual Costs — Development Period

The following figures are taken directly from Azure Cost Management, filtered to all resources associated with team-r2-2026, covering the project period from mid-May to 10 June 2026.

Azure Service	Category	Actual Cost	Notes
Virtual Machines	Compute	€67.29	Azure ML training jobs, sweep, pipeline runs
Azure Container Apps	Compute	€0.97	Frontend + API backend serving
Load Balancer	Deployment	€8.52	Kubernetes cluster networking
Storage	Deployment	€4.83	Datasets, model versions, MLflow artefacts
Virtual Network	Deployment	€1.56	Workspace networking
Bandwidth	Deployment	~€0.02	Data egress
Total		€83.19	Apr 13 – Jun 10, 2026

Breaking down by individual resource:

Resource	Actual Cost	Description
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team-r2-2026 (ML workspace)	€82.21	All training compute, storage, networking
signlang-r2-frontend (Container App)	€0.54	Vue 3 frontend serving
signlang-r2-api (Container App)	€0.43	FastAPI inference backend
Total	€83.19	

Of the €83.19 total, €82.21 (98.8%) is attributed to the Azure ML workspace covering training activity, and only €0.97 (1.2%) to live inference serving. This confirms that compute for model development dominates project costs, while deployment infrastructure is cost-efficient at current scale.

3. Compute Cost Analysis

3.1 Training Compute — Azure ML

Training compute is billed per second of VM usage on the attached Kubernetes cluster. Over the project period, 79 jobs were submitted across 9 experiments, including one full hyperparameter sweep (initially 12 trials for model registration testing, reduced to 6 trials). The Virtual Machine cost of €67.29 covers all of this activity.

Experiment type	Approx. cost	Notes
Single training jobs (submit_training_job.py)	Low	~2–9 min per run; most cost from failed retries
Full pipeline (submit_pipeline.py)	Medium	Preprocessing + augmentation + training stages
Hyperparameter sweep (submit_sweep_job.py)	High	6 trials, median early stopping; 12 trials used briefly for initial model registration testing
Smoke tests (smoke_test.py)	Negligible	Short validation jobs, cpu-small instance type

In production, training activity reduces to periodic retraining runs triggered by `check_training_triggers.py` — either monthly or when 100 new images are added to the dataset. This smart trigger avoids unnecessary runs, keeping ongoing compute cost proportional to actual data growth rather than the passage of time.

Using an Azure ML Standard_DS3_v2 VM (~€0.19/hour) for a 2-hour monthly retraining run, the projected steady-state training compute cost is approximately €0.76–€1.52/month — a negligible figure compared to development-period costs. The current setup uses the BUas on-premise Kubernetes cluster (lambda-2), making training cost €0 in Azure terms.

3.2 Inference Compute — Azure Container Apps

Inference is served by the signlang-r2-api Container App on the Consumption plan. This plan bills per vCPU-second and GiB-second of active usage, with a generous free tier (180,000 vCPU-seconds and 2 million requests per month). Scale-to-zero means no compute cost when no users are active.

The Azure Pricing Calculator was used to project inference compute costs at higher traffic levels. Configuration used: 1 vCPU, 2 GiB RAM, Consumption plan, West Europe region, scale-to-zero enabled.

Billing dimension	Unit price	Phase 1 (20 sessions/day)	Phase 2 (200 sessions/day)	Source
vCPU-seconds (above free tier)	\$0.000024/s	~€4.30	~€17.30	Pricing calculator
Memory GiB-seconds (above 360K free/month)	\$0.000003/GiB-s	~€1.10	~€4.30	Pricing calculator
Requests (above 2M free/month)	\$0.40/1M	~€0.64	~€13.60	Pricing calculator
Observed (current low traffic)	—	€0.43 / 4 weeks	—	Azure Cost Management
Projected monthly total		~€6.04	~€35.20	

The observed \$0.43 over four weeks at current low traffic validates the Phase 1 calculator figure of \$1.12/month. Both the pricing calculator and Azure Cost Management confirm inference costs are minimal at this scale. The free tier absorbs most cost at low scale — the 180,000 free vCPU-seconds covers approximately 20 sessions/day before billable usage begins.

3.3 CPU vs GPU Trade-off

A GPU-backed container would reduce per-frame inference latency but at significantly higher cost. EfficientNet-B0 is a compact model (~5M parameters) and CPU inference at 10 fps is achievable within the ~150ms WebSocket response window required for real-time fingerspelling recognition. CPU was therefore chosen as the cost-effective default. The actual €0.43/month observed cost confirms this is viable at current scale. GPU remains a future option if concurrent load testing reveals latency issues.

4. Deployment Cost Analysis

Deployment costs cover the infrastructure required to deliver SignSee to users: the Container Apps environment, CI/CD pipeline execution, storage, networking, and model serving infrastructure.

4.1 Container Apps Environment

Both the frontend and API run inside a single Azure Container Apps Environment. The Consumption plan environment has no fixed monthly fee — costs are purely usage-based (vCPU-seconds and requests). This is confirmed by the observed €0.97 total across both apps over four weeks of deployment.

The frontend (signlang-r2-frontend) serves the compiled Vue 3 static build and costs €0.54 over the period. The API (signlang-r2-api) serves WebSocket inference requests and costs €0.43. Both are deployed in the North Europe region (Azure Container Apps Environment: cae-y2d-2026-r2).

4.2 CI/CD Pipeline — GitHub Actions

Deployments are automated via GitHub Actions workflows covering: automated tests, Docker image build and push to GitHub Container Registry (GHCR), deployment to Portainer (on-premise), and deployment to Azure Container Apps. GitHub Actions provides 2,000 free minutes per month on public repositories.

Workflow	Est. duration	Monthly runs	Monthly minutes used
Test suite + lint (CI)	~3 min	~20 (per PR/push)	~60 min
Docker build + push to GHCR	~5 min	~10 (per merge)	~50 min
Deploy to Azure Container Apps	~3 min	~10 (per merge)	~30 min
Total estimated monthly usage			~140 min / 2,000 free

Estimated usage of ~140 minutes/month is well within the 2,000-minute free tier, making CI/CD pipeline execution cost €0.00. This assumes no GPU runners or self-hosted runners are used.

4.3 Container Registry — GitHub Container Registry

Docker images for the frontend and API are stored in GitHub Container Registry (GHCR) rather than Azure Container Registry (ACR). GHCR is free for public repositories and included in GitHub Pro/Teams for private repositories. This avoids the ACR Basic tier flat fee of approximately €4.70/month. The project already uses GHCR in its CI/CD pipeline, so no migration is required.

4.4 Model Serving & Blue-Green Deployment

The model serving infrastructure uses an Azure ML online endpoint deployed to the attached lambda-2 Kubernetes cluster. Deployment uses a canary/blue-green strategy implemented in `rollout_canary.py` and `rollout_blue_green.py`, which temporarily runs two deployments in parallel during rollouts. On the BUas on-premise cluster (lambda-2) this incurs no additional Azure cost. In a fully independent Azure deployment using Azure ML Managed Online Endpoints instead of Container Apps, costs would be significantly higher: managed online endpoints bill at approximately \$0.20/hour per instance continuously while the endpoint is running, regardless of whether requests are being served. At 24/7 uptime that equates to approximately \$144/month per instance — making Container Apps the far more cost-effective choice for SignSee's usage pattern.

4.5 Storage

Azure Blob Storage (workspaceblobstore) holds all datasets and model artefacts. Actual cost of €4.83 covers the full development period including 7 registered data assets and 12+ model versions. In steady-state production, storage grows by approximately one model version (~50–100 MB) per retraining run, keeping ongoing costs near €1–2/month.

4.6 Networking

Networking costs (Load Balancer €8.52, Virtual Network €1.56) are attributable to the Kubernetes cluster infrastructure shared across BUas teams — these are not direct SignSee deployment costs in production. In a standalone Azure deployment using Container Apps only, there is no Load Balancer or Virtual Network charge under the Consumption plan. Outbound bandwidth is free up to 100 GB/month, which is not approached at current traffic levels.

5. Full Production Cost Summary

Combining compute and deployment costs across both phases:

Component	Category	Phase 1	Phase 2	Notes
Inference API (Container App)	Compute	~€6.04	~€35.20	Scale-to-zero, observed baseline €0.43/mo
ML Retraining	Compute	€0.00	€0.00	BUas on-prem; €0.76–1.52 if Azure VM
Frontend (Container App)	Deployment	~€0.54	~€2.00	Observed baseline €0.54/mo
CI/CD (GitHub Actions)	Deployment	€0.00	€0.00	Within 2,000 min free tier
Container Registry (GHCR)	Deployment	€0.00	€0.00	Free; replaces ACR (saves €4.70/mo)
Storage (Blob)	Deployment	~€1.00	~€1.50	Observed €4.83 over full dev period
Networking / Load Balancer	Deployment	€0.00	€0.00	Not applicable in standalone Container Apps
Estimated Monthly Total		~€7.58	~€38.70	

6. Performance vs Cost Trade-offs

Scale-to-Zero vs Always-On

The Consumption plan with scale-to-zero avoids a fixed monthly minimum charge, keeping cost proportional to actual usage. The trade-off is a cold-start delay of 1–3 seconds on the first request after idle. At Phase 1 scale with intermittent usage, this is acceptable. At Phase 2 with consistent traffic, enabling a minimum replica would add ~€40–50/month but eliminate cold starts entirely.

Smart Retraining Trigger

The `check_training_triggers.py` script evaluates data changes first (minimum 100 new images) before falling back to the time-based monthly schedule. This prevents unnecessary training runs when no meaningful new data is available, keeping retraining compute proportional to actual data growth.

On-Premise vs Cloud Training

Running training on the attached BUas Kubernetes cluster (lambda-2) rather than an Azure-provisioned VM cluster eliminates all VM compute cost for training. If the project migrates to a fully independent deployment, a `Standard_DS3_v2` cluster with `min_instances = 0` would cost approximately €0.76–€1.52/month for monthly 2-hour retraining runs — still a minor cost relative to inference at scale.

7. Cost Control Strategy

- Scale-to-zero on both Container Apps — compute charged only during active sessions.
- Smart retraining trigger — avoids unnecessary Azure ML job submissions.
- GHCR over ACR — saves ~€4.70/month with no functional trade-off.
- CI/CD within GitHub Actions free tier — no pipeline execution costs at current commit frequency.
- Azure Cost Management budget alert — a €50/month cap is configured on the BUas subscription; an equivalent alert should be set for any independent production deployment.
- Monitoring dashboard (in deployment) — tracks per-request latency and error rates to detect runaway retry loops or misconfigured autoscaling before costs escalate.

8. Summary

Figure	Value
Total actual Azure spend (Apr–Jun 2026)	€83.19
Of which: training compute	€67.29 (80.9%)
Of which: deployment infrastructure	€15.47 (18.6%)
Of which: live inference serving	€0.97 (1.2%)
Projected production cost — Phase 1	~€7.58 / month
Projected production cost — Phase 2	~€38.70 / month
CI/CD pipeline cost	€0.00 (within free tier)
Container registry cost	€0.00 (GHCR)
Training compute in production	€0.00 (BUas on-prem Kubernetes)